

Figure 1: Function

example, if you want help on *function random*, type ‘help rand’ without quotes in the command window.

Why Scilab then?

Scilab was created in 1990 by researchers from Inria and ENPC in France. It is freely available for all computation platforms. Scilab stores numbers in IEEE 754 double-precision binary floating-point format using 64 bits or 8 bytes of computer memory, which is also 16 decimal digit accuracy.

It has computation for elementary mathematics, linear algebra, numerical integration, differentiation, optimisation, etc. Its interface is similar to MATLAB's and includes a workspace, editor, command window, working directory, etc. It has *xcos*, which is equivalent to *simulink* in MATLAB. Since it doesn't have as many inbuilt functions and tool bars as in MATLAB, many find it less interesting to work with Scilab. However, real researchers, who build from scratch, find Scilab a useful platform. But engineers whose problems are too complex to solve in an exact manner, prefer MATLAB. Engineering research employs many semi-empirical methods that are foreign to pure scientific research—one example being the method of parameter variation. Such methods can be easily implemented in Scilab.

Fire-fly algorithm in Scilab

To feel the power of Scilab, let's implement a swarm intelligence based algorithm in it. Note that we use a single variable for ease of visualisation. The function $f(x) = x \sin(x)$ has a single variable x , and we want to find the value for x such that the function is minimum. Essentially, the above example is a minimisation problem, and these kinds of problems come under optimisation. If we plot the function for variable x , by looking into the function plot, we can infer that the minimum value lies somewhere between 17 and 18. To find it with the fire-fly algorithm, we have the help of 25 fire-flies. We scatter the flies along the function, so that we have flies in different locations along the function plot. The brightness of the flashes given out by each fly increases

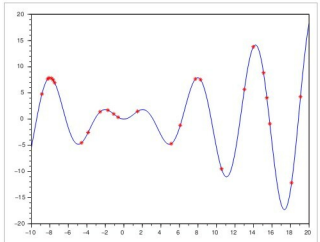


Figure 2: Function with flies

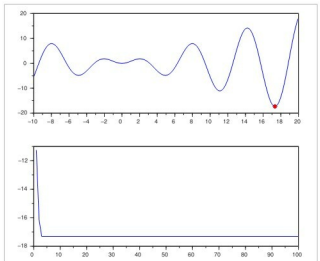


Figure 3: Function convergence

with its proximity to the exact location of the solution. The equation governing this process is:

$$x_i = x_i + \beta_j \times e^{a_j - a_i} \times (x_j - x_i) + \alpha \epsilon_i$$

After 100 iterations, the result obtained is 17.307. This is shown with the convergence, as seen in Figures 1, 2 and 3. Hence, we have seen the action of the fire-fly algorithm in finding the minimum, using Scilab.

Engineering problems, which are more like real world problems, tend to be highly non-linear in nature. They can be easily tackled with evolutionary methods like this. Unfortunately, Scilab is mostly ignored by academia, and its power is often overlooked. **END**

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